

COMS BC1016

Introduction to Computational Thinking and Data Science

Lecture 5: Histograms

BARNARD COLLEGE OF COLUMBIA UNIVERSITY

Sept 30, 2025

**Last Time: Line Plots, Scatter
Plots, and Bar Charts**

Attribute Types

- Table columns are attributes
- All values in a column should be the same type *and* comparable to each other
 - **Numerical:** Values are on a numerical scale (e.g., years)
 - Numerical values are ordered
 - Differences are meaningful
 - **Categorical:** Each value is from a fixed inventory (e.g., material)
 - May or may not have an ordering
 - Categories are either the same or different

Is this a numerical or categorical attribute?

Speed Skating: The **time** to skate 3000m (in min and sec)

[3:54, 3:56, 3:59, 4:01]

Is this a numerical or categorical attribute?

Skating: **Events that figure skaters can perform in**

['single skating', 'pair skating', 'ice dance', 'team event']

Is this a numerical or categorical attribute?

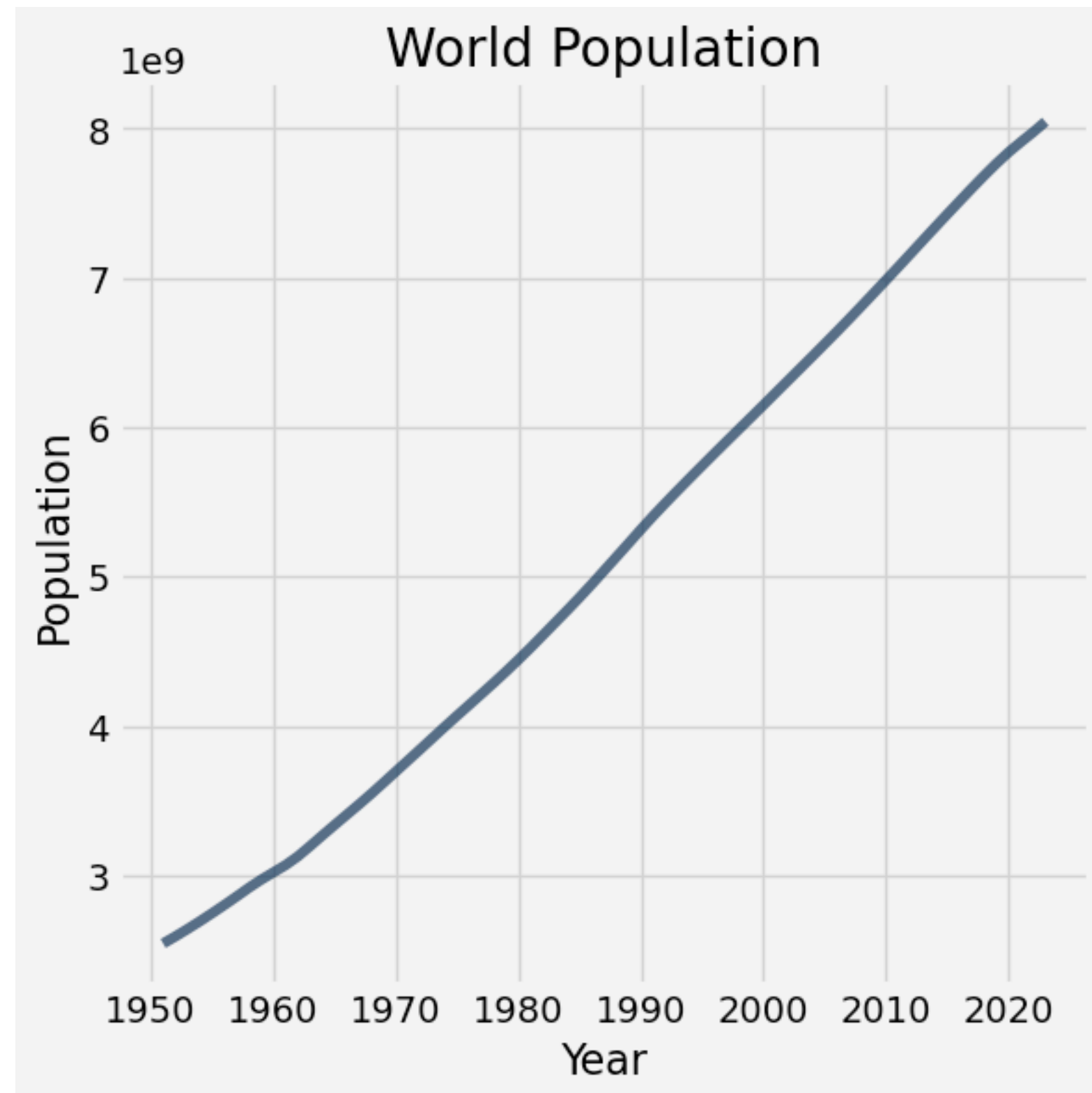
Speed Skating: **Race distances (in meters)**

[500, 1000, 1500, 3000, 5000]

Plotting Two Numerical Variables

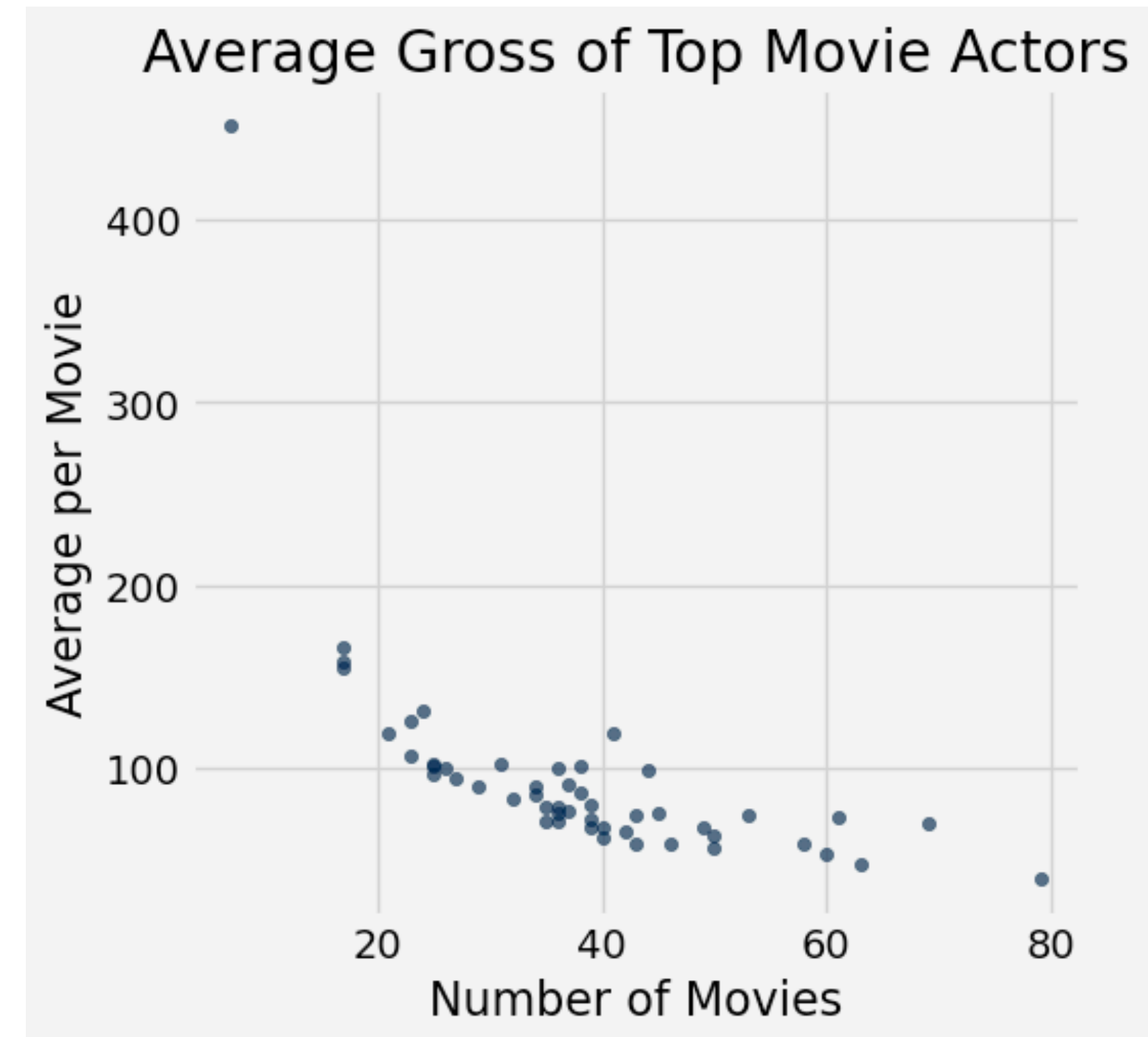
Line Plot

`plot`



Scatter Plot

`scatter`



Line Plots vs Scatter Plots

```
tbl.plot(x_label, y_label)
```

```
tbl.scatter(x_label, y_label)
```

- Use **line plots** for **sequential** data if:
 - x-axis has an order (e.g., time, years, distance)
 - sequential differences in y value are meaningful
 - there's only one y-value for each x-value
- Use **scatter plots** for **non-sequential** quantitative data if:
 - looking for associations

Demo: Practice with Plots

Bar Charts for Categorical Attributes

```
tbl.barh(category_label, freq_label)
```

Bar charts are good for visualizing:

- relationships between categorical variable(s) and a numerical variable

Charts (So Far)

Type	Syntax	Description
Line graph	<code>.plot(x_axis, y_axis)</code>	Sequential numerical data
Scatter Plot	<code>.scatter(x_axis, y_axis)</code>	Non-sequential numerical data
Bar Chart	<code>.barh(column_label)</code>	Relationship of categorical variable(s) and numerical values

Charts in this Class

Type	Syntax	Description
Line graph	<code>.plot(x_axis, y_axis)</code>	Sequential numerical data
Scatter Plot	<code>.scatter(x_axis, y_axis)</code>	Non-sequential numerical data
Bar Chart	<code>.barh(column_label)</code>	Relationship of categorical variable(s) and numerical values
Histogram		Distribution of numerical data

Survey!

- <https://forms.gle/tASMYkYq7FDpYtkQ8>
- Please take a few moments to answer these questions for a demo later



Distributions

Terminology

- **Individuals**: those whose attributes are recorded (instances of data)
- **Variable**: an attribute or column
 - Can be numerical or categorical
 - Each **individual** has exactly **one value**
 - Has a **distribution**:
 - For each different value of a variable, there is a frequency of individuals that have that value

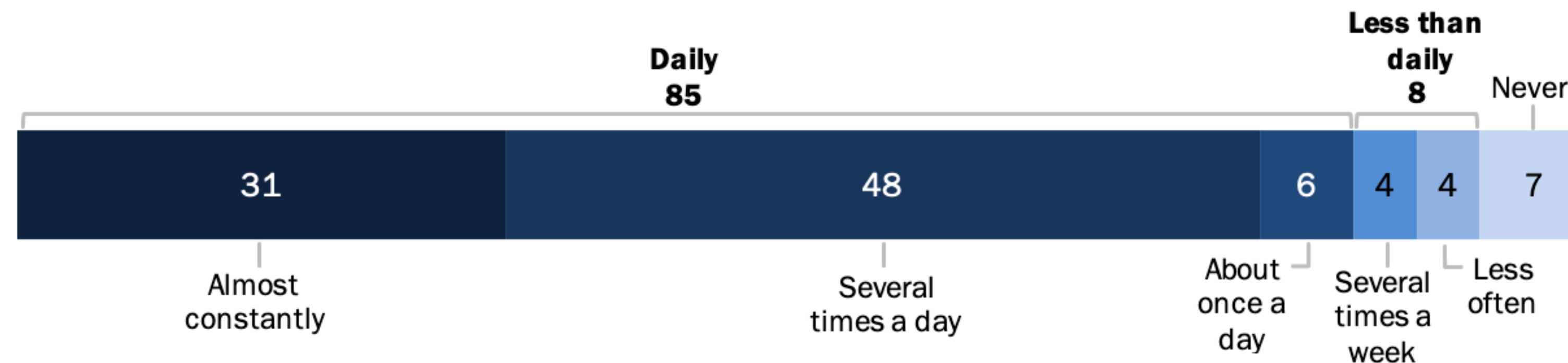
Distributions

Each individual has exactly one category

- Percents add to 100

More than eight-in-ten U.S. adults go online at least daily

% of U.S. adults who say they go online ...



Note: Respondents who did not give an answer are not shown.

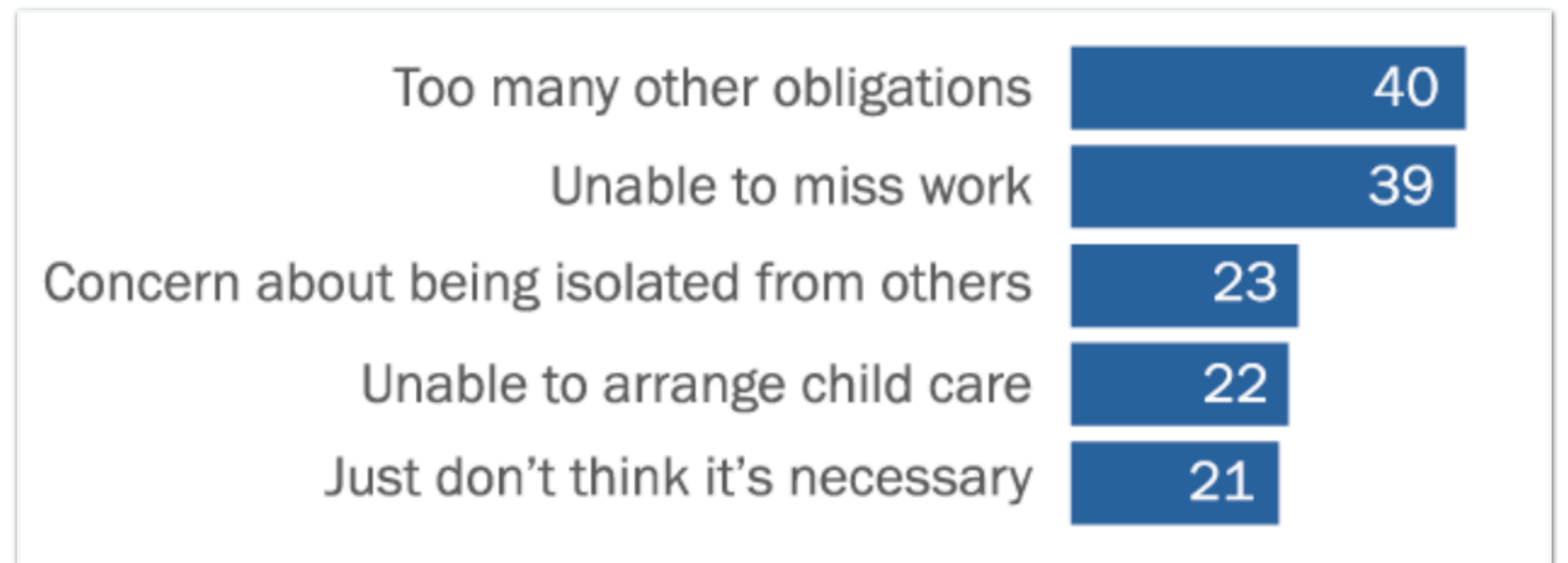
Source: Survey of U.S. adults conducted Jan. 25-Feb. 8, 2021.

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Not a distribution

Percents of survey respondents on “a major reason you would find it difficult to quarantine for 14 days”

Respondents can pick more than one answer



Bars in this graph represent overlapping groups.

Bar Charts for Categorical Distributions

To display **categorical variables** and their **frequencies**:

- Bar chart
 - One bar for each category
 - Ordering of the bar can be specified (e.g. `.sort`), but any order is fine
 - Length of bar is the percent (or count) of individuals in the category
- Grouping
 - Given a table of individuals, create a table of counts using the **group** method

Grouping Categorical Data: **group**

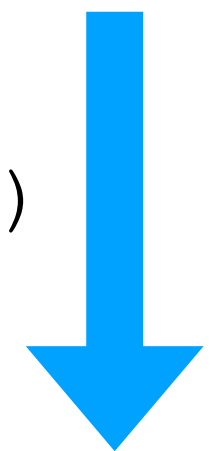
```
tbl.group(column_label)
```

group can be used to **count** the number of rows of **each category** in a column

- Can also be used with multiple columns and to aggregate attributes
- Will cover advanced usages next week

Name	Age	Weight	Coloring	Sex	Owner
Ruby	14	8	tuxedo	F	Alice
Gertrude	15	12	tuxedo	F	Alice
Hamby	8	16	tabby	M	Bob
Fig	3	7	tabby	F	Bob
Corina	6	10	tortie	F	Carol
Frito	2	8.5	tabby	M	Carol

```
cat_tbl.group('Coloring')
```



Coloring	count
tabby	3
tuxedo	2
tortie	1

DEMO

Grouping Numerical Values: Binning

Binning is counting the number of numerical values that lie within a range (called bins)

- Bins are defined by their lower bounds (inclusive)
- The upper bound is the lower bound for the next bin
 - Example: $[10, 20)$ includes 10 but does not include 20

Binning: Example

[1, 5, 7, 3, 2, 11, 18, 16, 15, 10, 22, 27, 4]

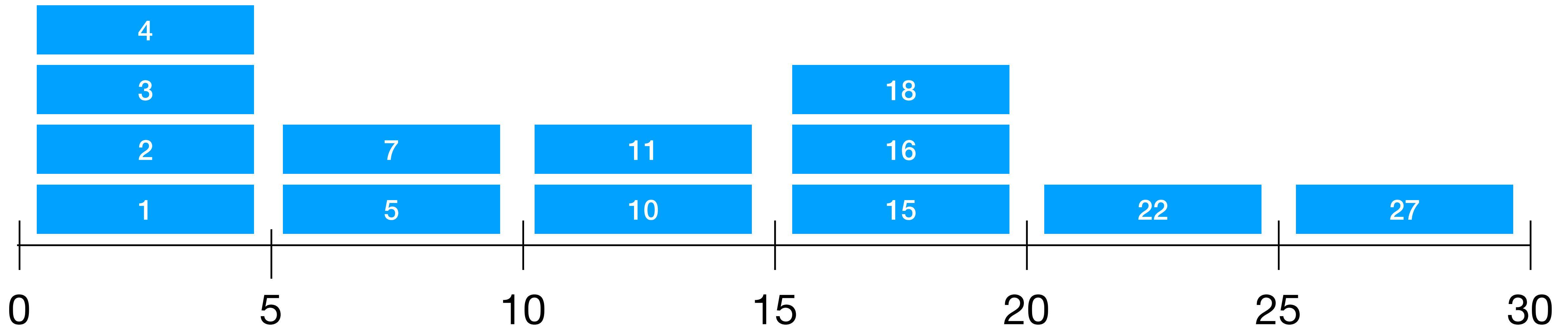
Let's say we decide to use a bin size of 5



Binning: Example

[1, 5, 7, 3, 2, 11, 18, 16, 15, 10, 22, 27, 4]

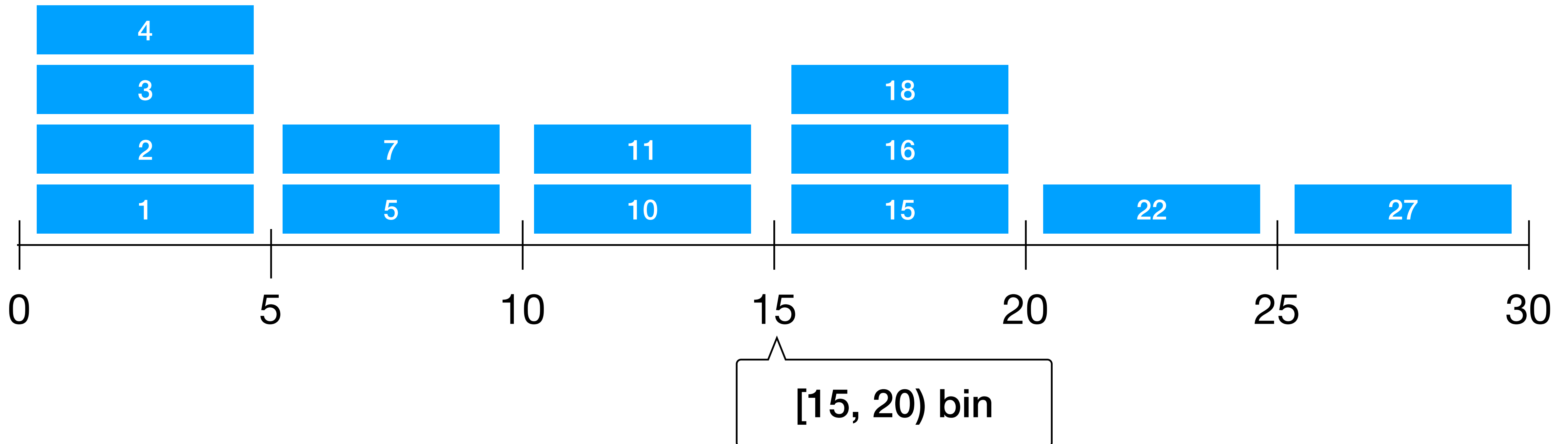
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Binning: Example

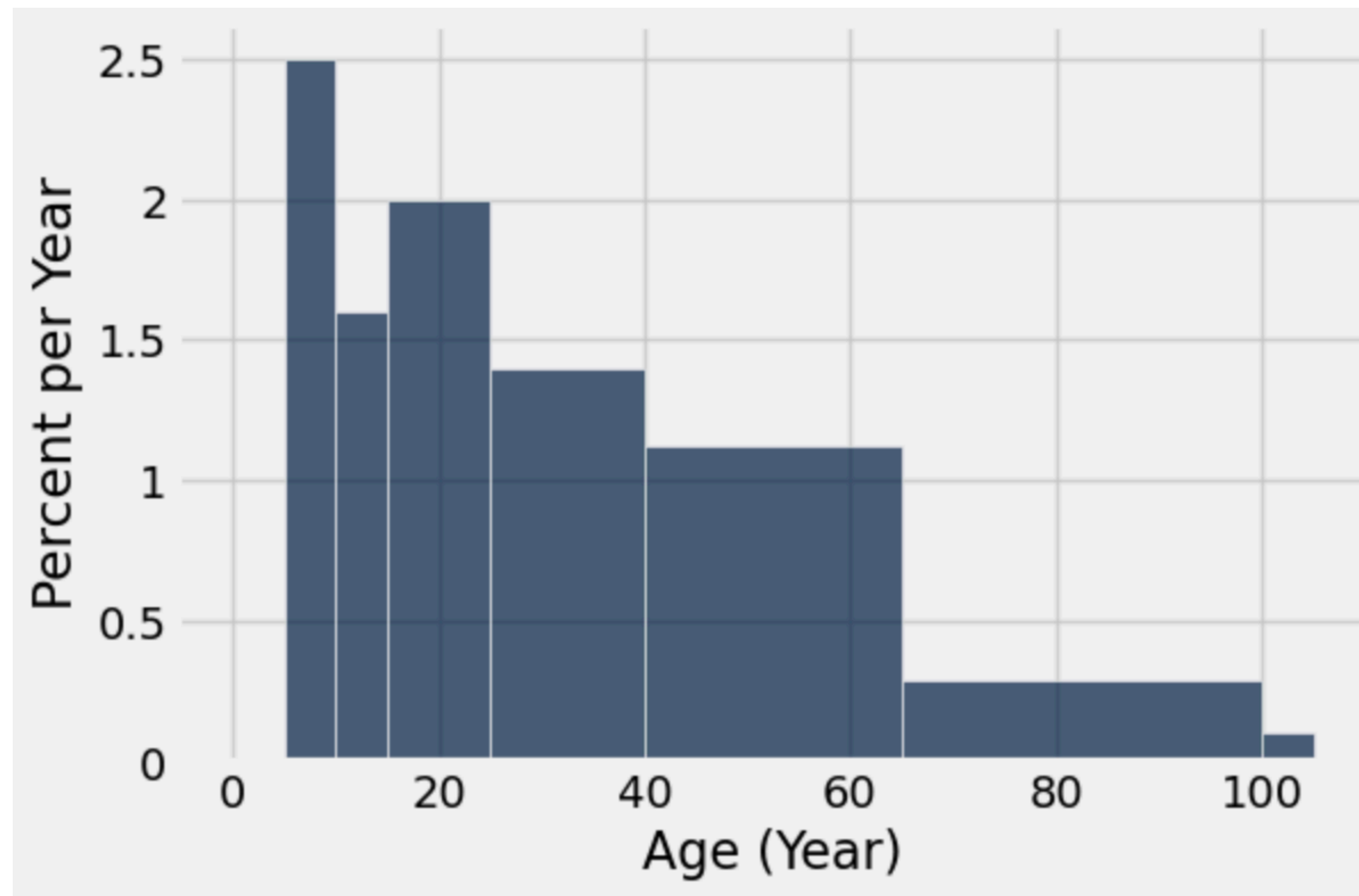
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Let's say we decide to use a bin size of 5



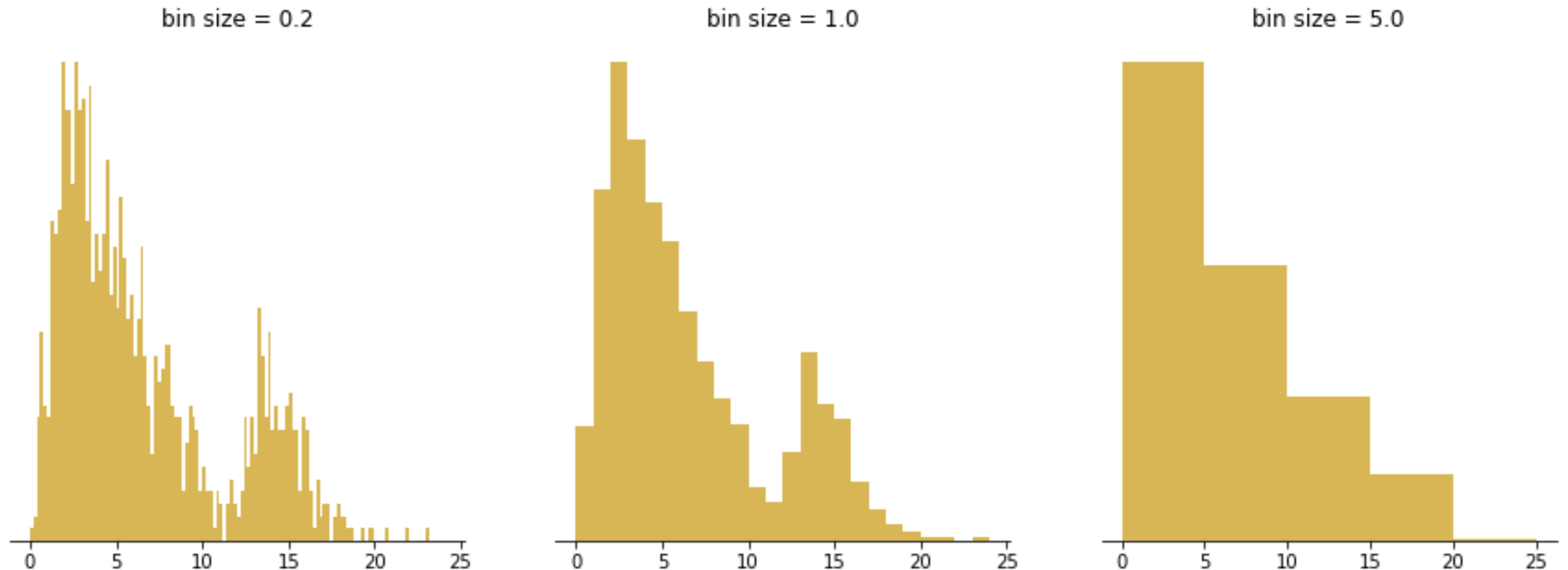
Plotting Numerical Distributions

- **Histograms** plot the distribution of a numerical variable using bins!



A Note on Bin Size

Choose so that it's representative of your data



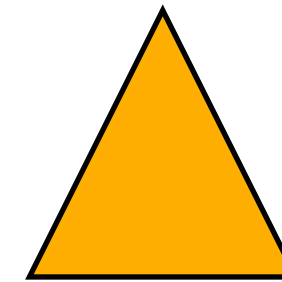
Area Principle

Area Principle

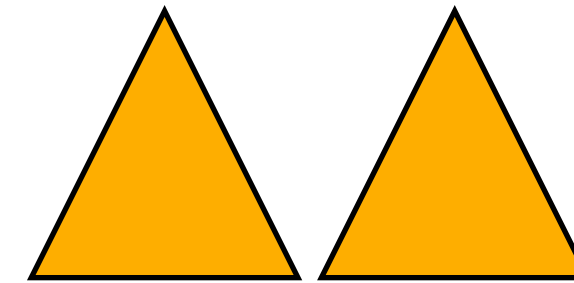
Areas should be *proportional* to the values they represent

For example

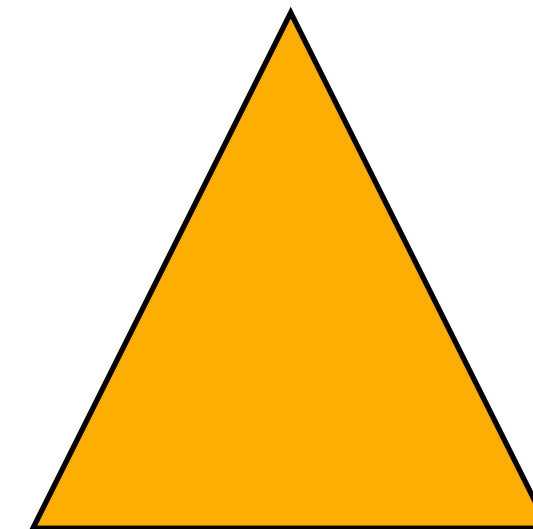
- If you represent 20% of a population by:



- Then 40% can be represented by:



- But not by:

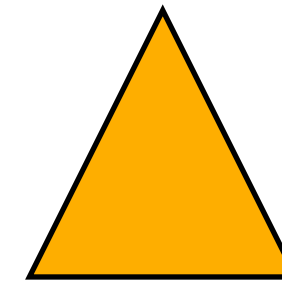


Area Principle

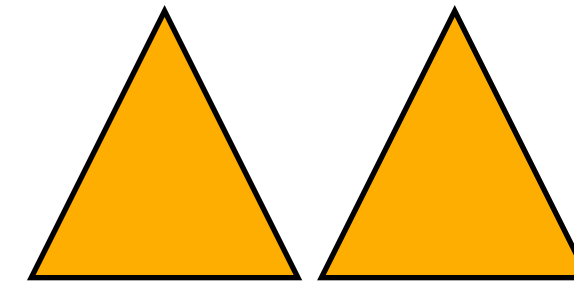
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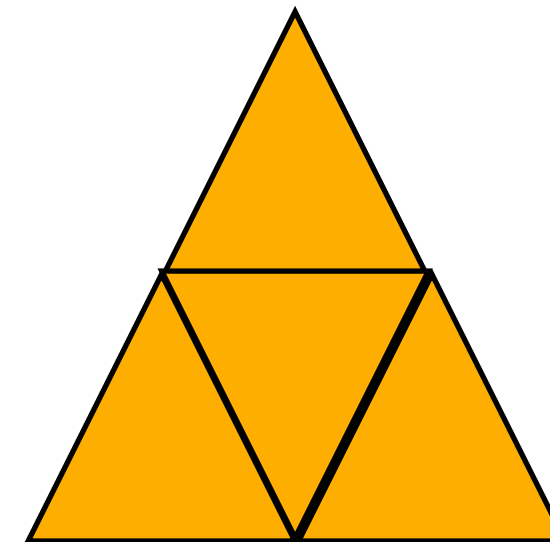
- If you represent 20% of a population by:



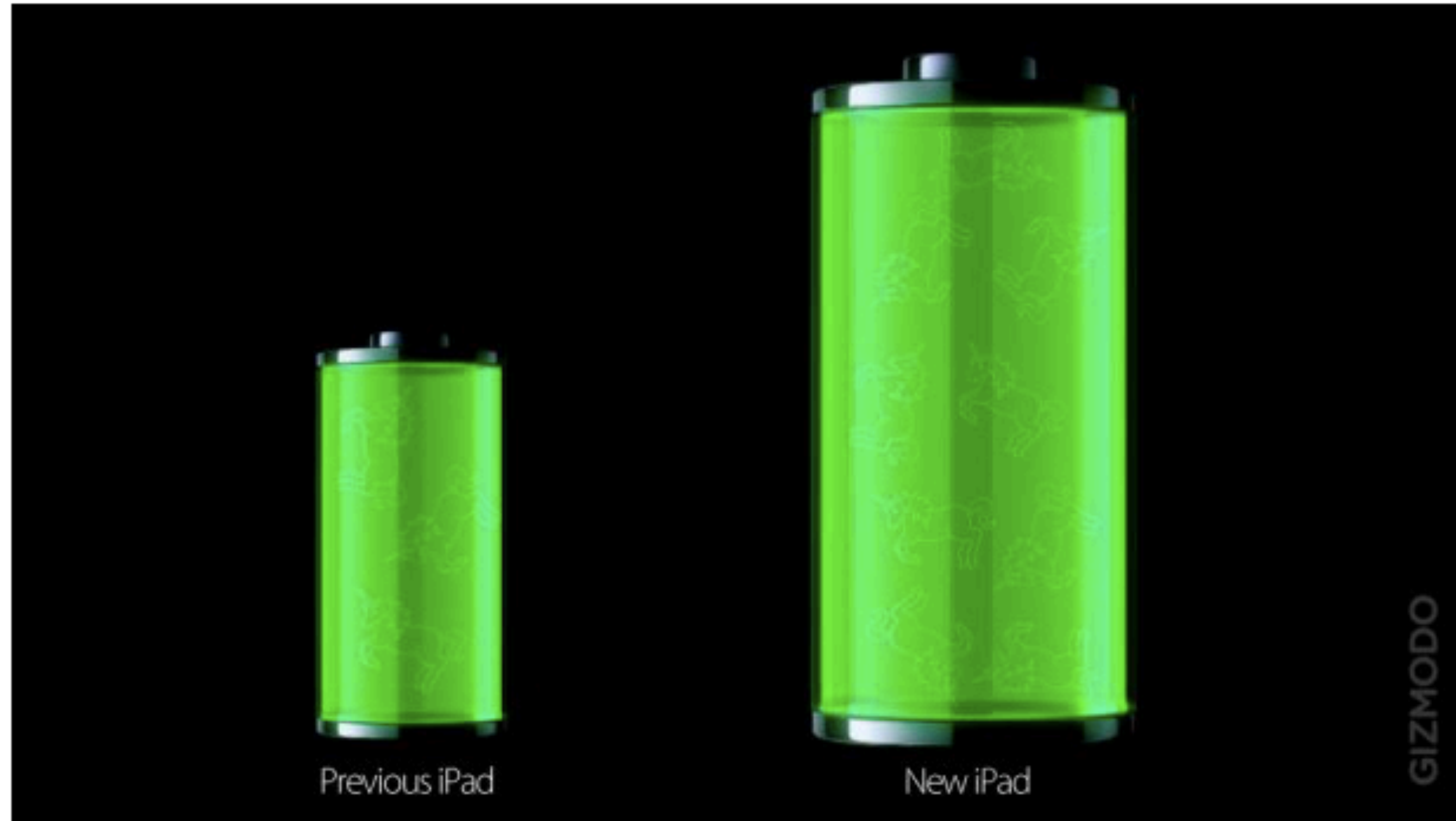
- Then 40% can be represented by:



- But not by:



Is the area principle being used here?

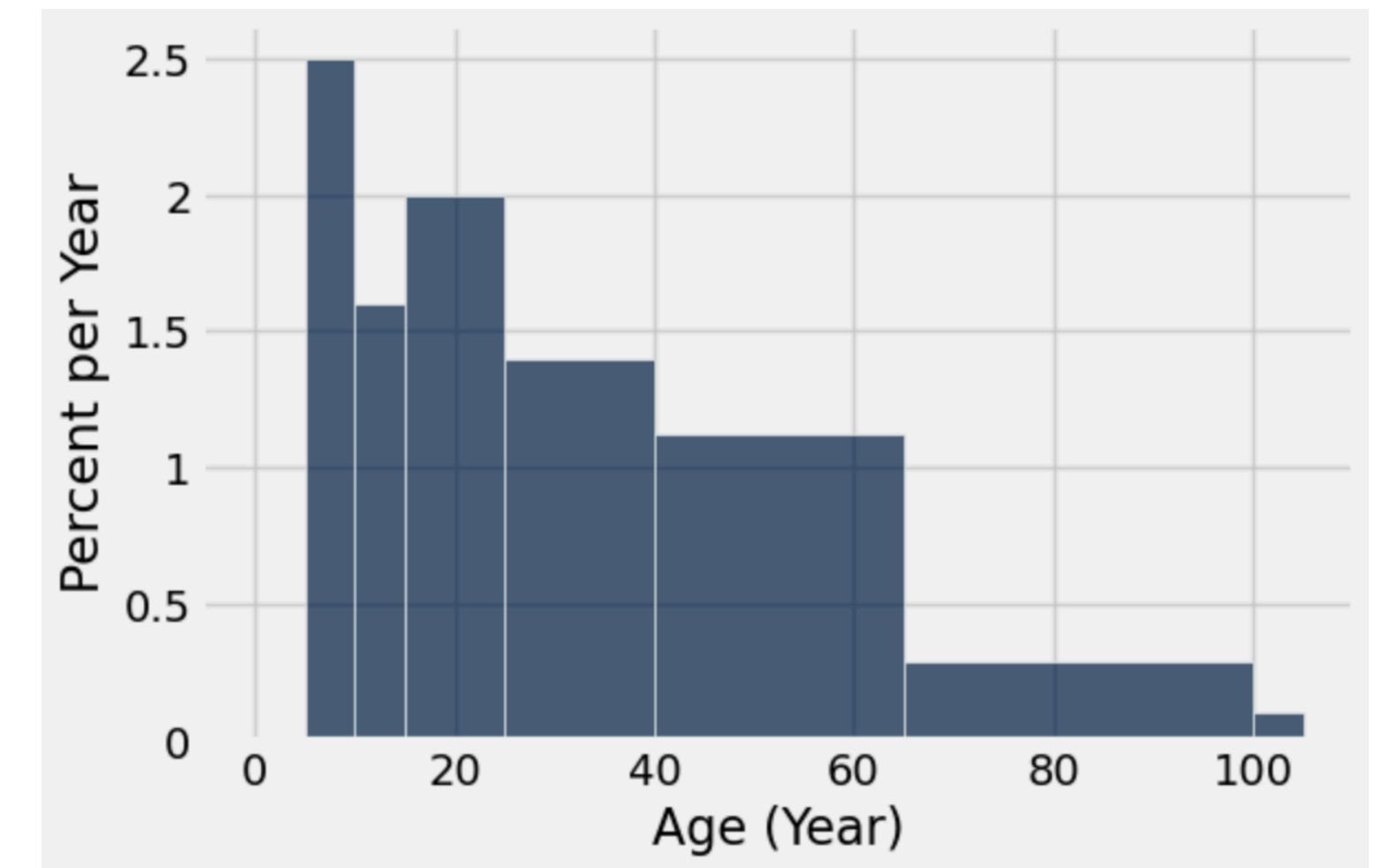


From [Gizmodo](#), this shows battery size in the new iPad versus that of the iPad 2. The battery in the former is 70 percent bigger than that of the latter. Something's not right here.

Histograms

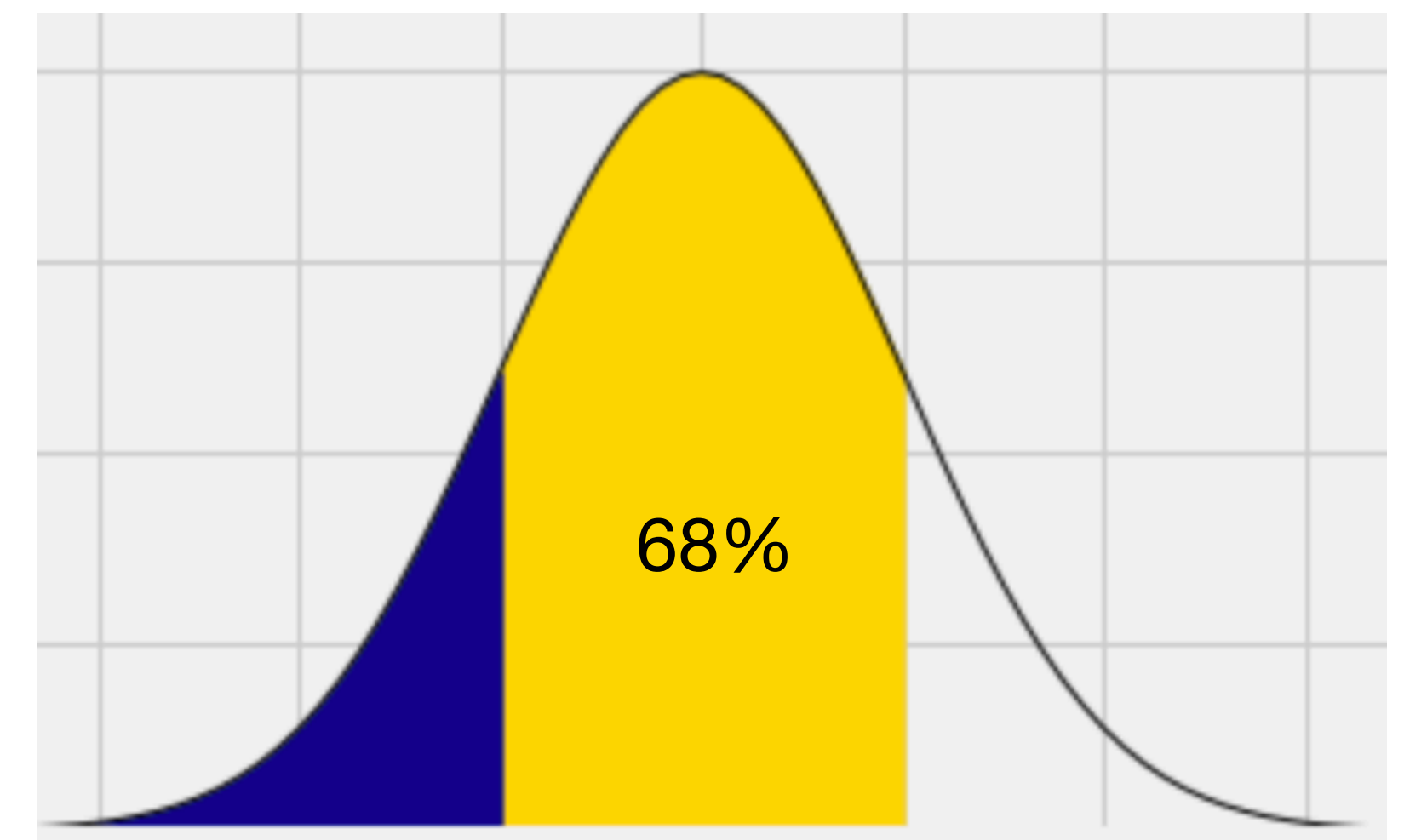
Histograms

- Displays the distribution of a numerical variable
- One bar corresponds to each bin
- Uses the area principle:
 - The **area** of each bar is the **percent** of individuals in the corresponding bin



Later in the course, we will approximate histograms with smooth curves.

Areas will still represent percents.



Demo: Histograms

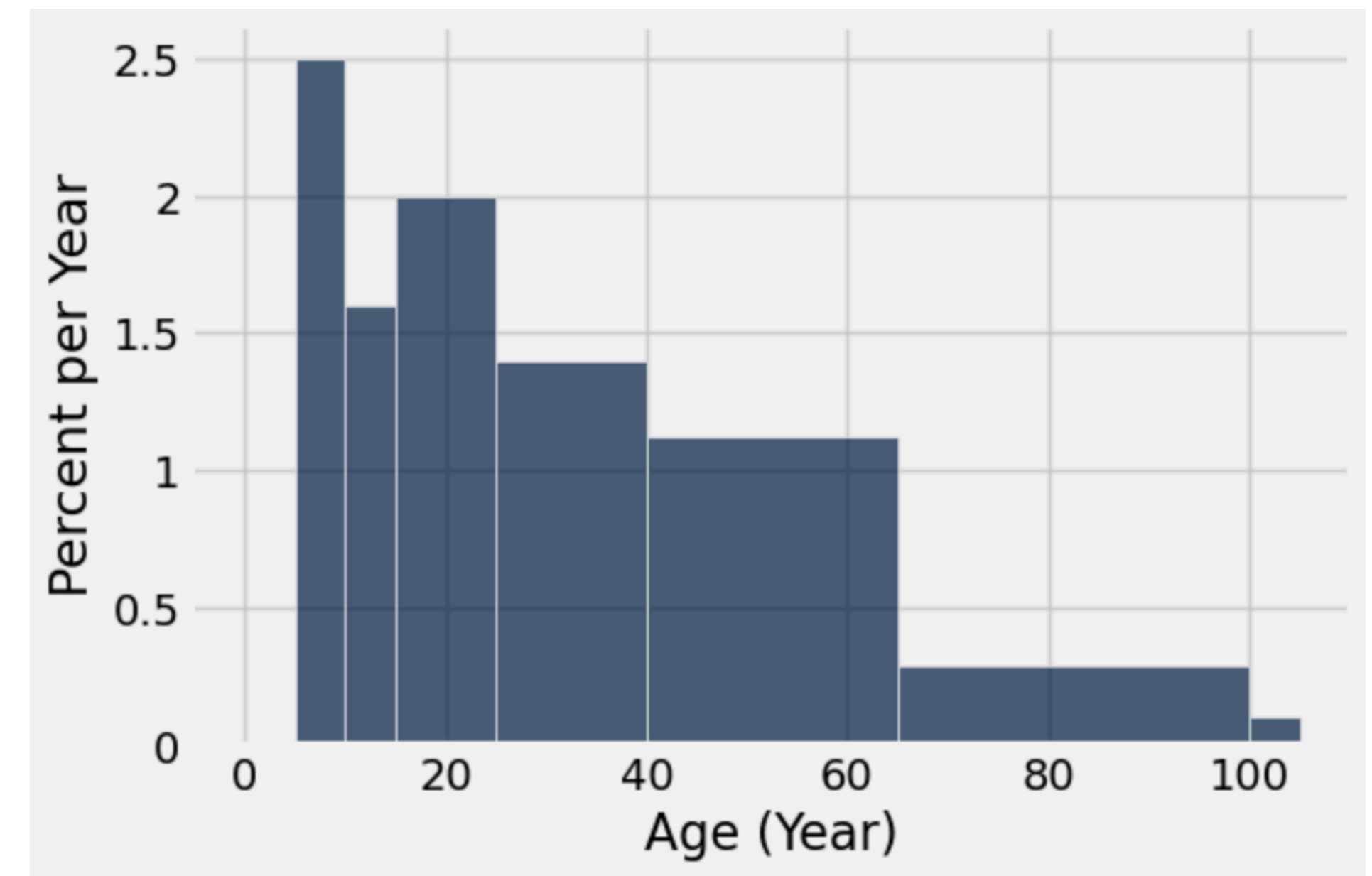
Histograms

By default, `hist` uses a scale that ensures the area of the chart sums to 100%

The **area** of each bar is a **percentage** of the whole

The **horizontal axis** is a number line (e.g., years), and bin sizes don't need to be equal

The **vertical axis** is a rate (e.g., percent/year), measuring the density

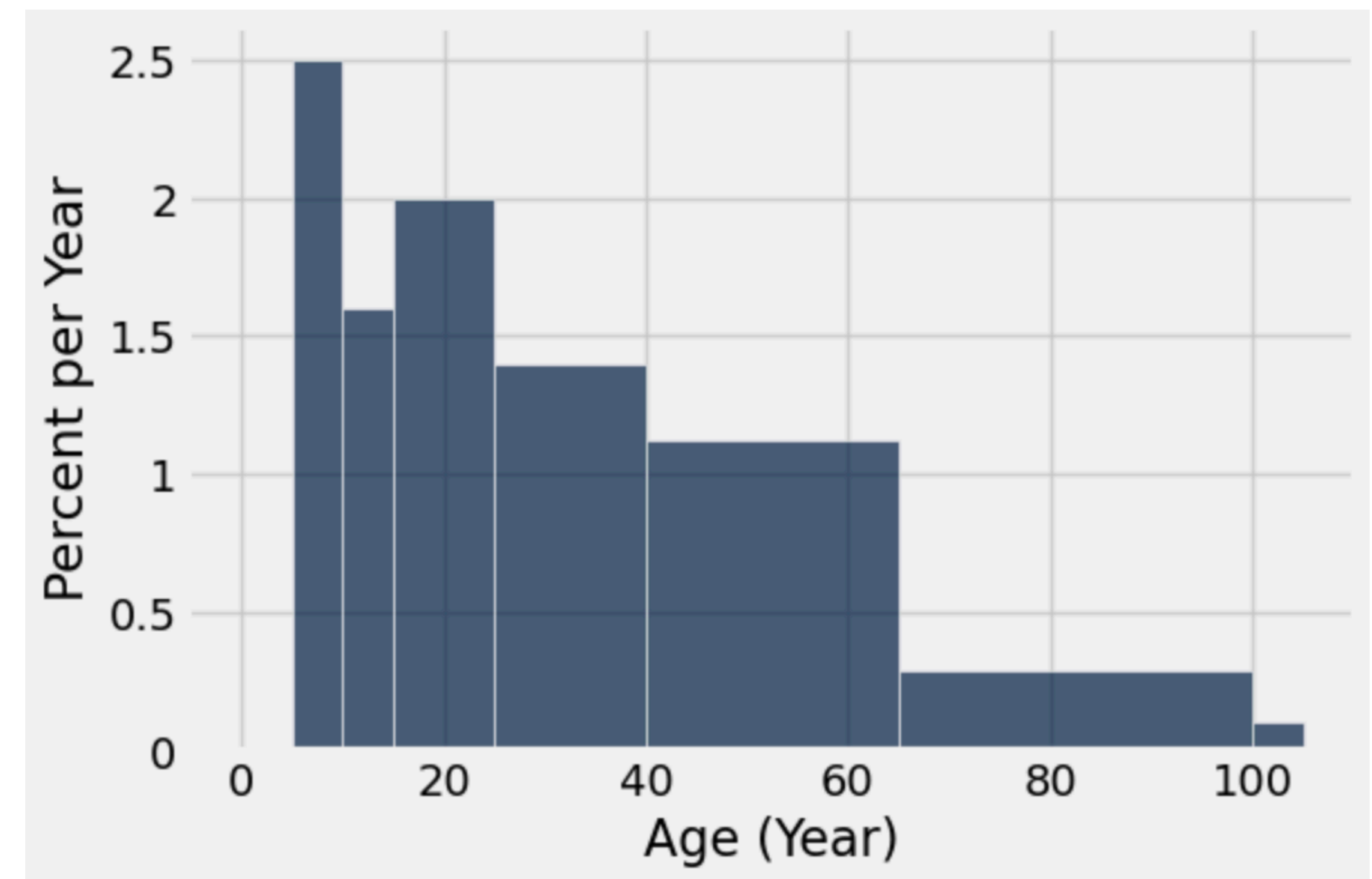


DEMO

Height Measures Density

- The height measures the percent of data in the bin *relative to the amount of space in the bin*
- Measures crowdedness or **density**
- Units: percent per x-axis unit

$$\text{height of bar} = \frac{\text{percent of entries in bin}}{\text{width of bin}}$$



How to Calculate Height

The [40, 65) bin contains 56 out of 200 movies

$$\text{height of bar} = \frac{\text{percent of entries in bin}}{\text{width of bin}}$$

How to Calculate Height

The [40, 65) bin contains 56 out of 200 movies

- What percentage is 56 out of 200?

$$56/200 = 28\%$$

- How wide is the [40, 65) bin?

$$65-40 = 25 \text{ years wide}$$

$$\text{height of bar} = \frac{\text{percent of entries in bin}}{\text{width of bin}}$$

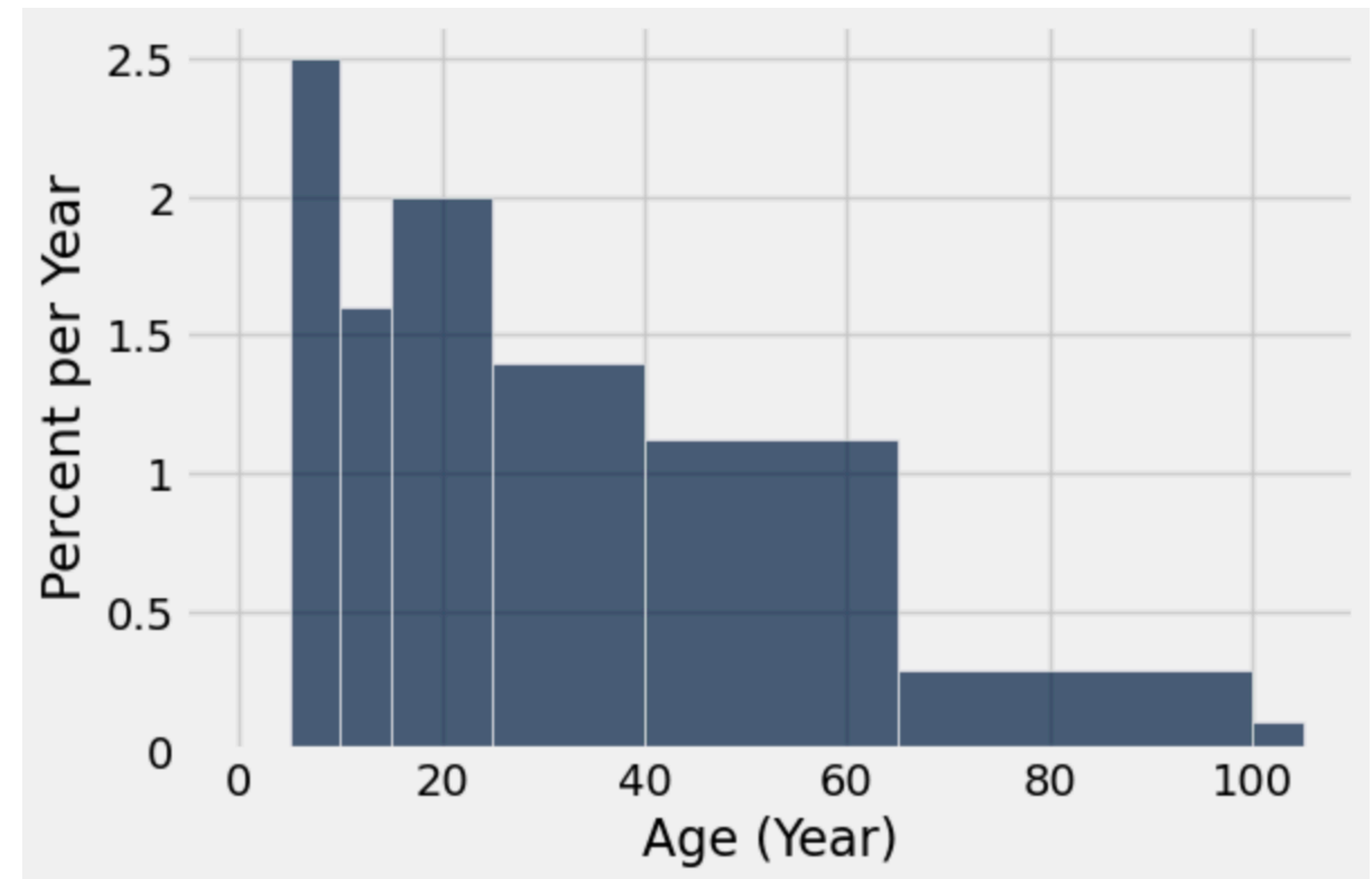
$$\begin{aligned} \text{height} &= \frac{\% \text{ in bin}}{\text{width of bin}} = \frac{28 \%}{25 \text{ years}} \\ &= 1.12 \% \text{ per year} \end{aligned}$$

Area Measures Percent

area of bar = percent of entries in bin
= (height of bar) $\times$ (width of bin)

When to use area vs height?

- “How many individuals in the bin?”
 - Use area
- “How crowded is the bin?”
 - Use height



Demo: More Histograms

Next Class

- Today
 - Visualizing categorical distributions (Bar chart)
 - Visualizing numerical distributions (Histograms)
- Monday
 - Functions and Groups